

Emotion-Aware Chat/Voice Assistant Using Face and Speech Recognition

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Computer Engineering



Engineering Success

Introduction

Human communication depends heavily on emotion, but most AI assistants focus only on words. This project explores whether an AI system can recognize emotional signals from human face and voice, then adapt its responses in a more supportive and natural way. Neural networks do not feel emotions, but they can learn patterns that correlate with emotional states from large labeled datasets. This research studies how well that learning works in real interactions. This project uses NVIDIA Jetson Orin Nano for local emotion sensing and Qwen3-4B-Instruct-2507 as the conversational AI assistant. The assistant takes emotional labels, confidence scores, and uncertainty signals to guide dialogue policy. More specifically, the functionalities of the final emotion-aware AI assistant include:

1. Face detection and tracking, followed by a compact CNN for categorical facial-expression recognition.
2. Emotion-conditioned dialogue generation with the Qwen3 model.
3. Natural speech output using a TTS pipeline.
4. For speech emotion (planned/in progress), we extract openSMILE features from the RAVDESS corpus and train a lightweight classifier for vocal emotion.

Goals:

1. Ensure user privacy by running video and voice data locally on Jetson/edge infrastructure instead of cloud-only processing.
 2. Build a multimodal conversational assistant that perceives a user's emotional state and adapts its responses in real time.
- Current conversational AI systems typically treat interaction as text-only, requiring users to self-regulate communication style when they are stressed or emotionally distressed. This project proposes an integrated emotion-aware assistant that combines facial affect recognition, speech/voice interaction, and emotion-conditioned dialogue generation in one workflow. Rather than relying on users to manually adjust how they communicate, the system automatically adapts response tone using detected emotion signals and confidence estimates. This design aims to reduce user burden and improve perceived empathy and naturalness in real-time interaction while preserving privacy through local processing.

Methodology

Facial Affect Sensing

Model

- Backbone: MobileNetV3 (compact CNN for edge-friendly inference)
- Classes: angry, disgusted, fearful, happy, neutral, sad, surprised
- Training objective: weighted cross-entropy with label smoothing (for class imbalance and robustness)

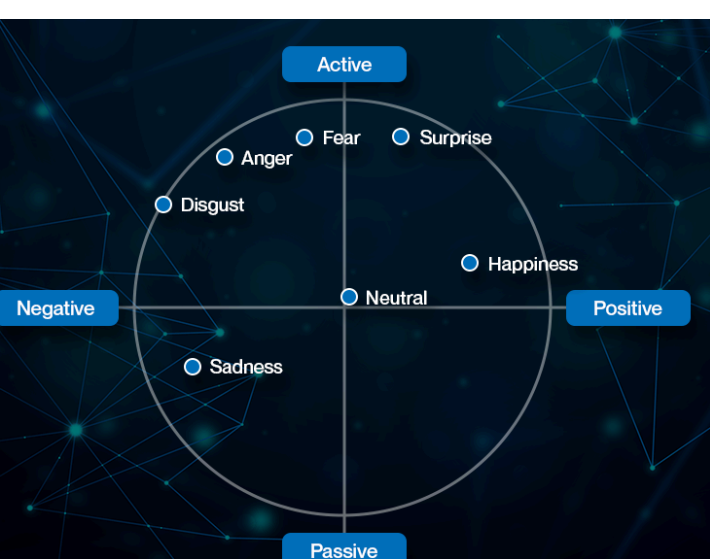
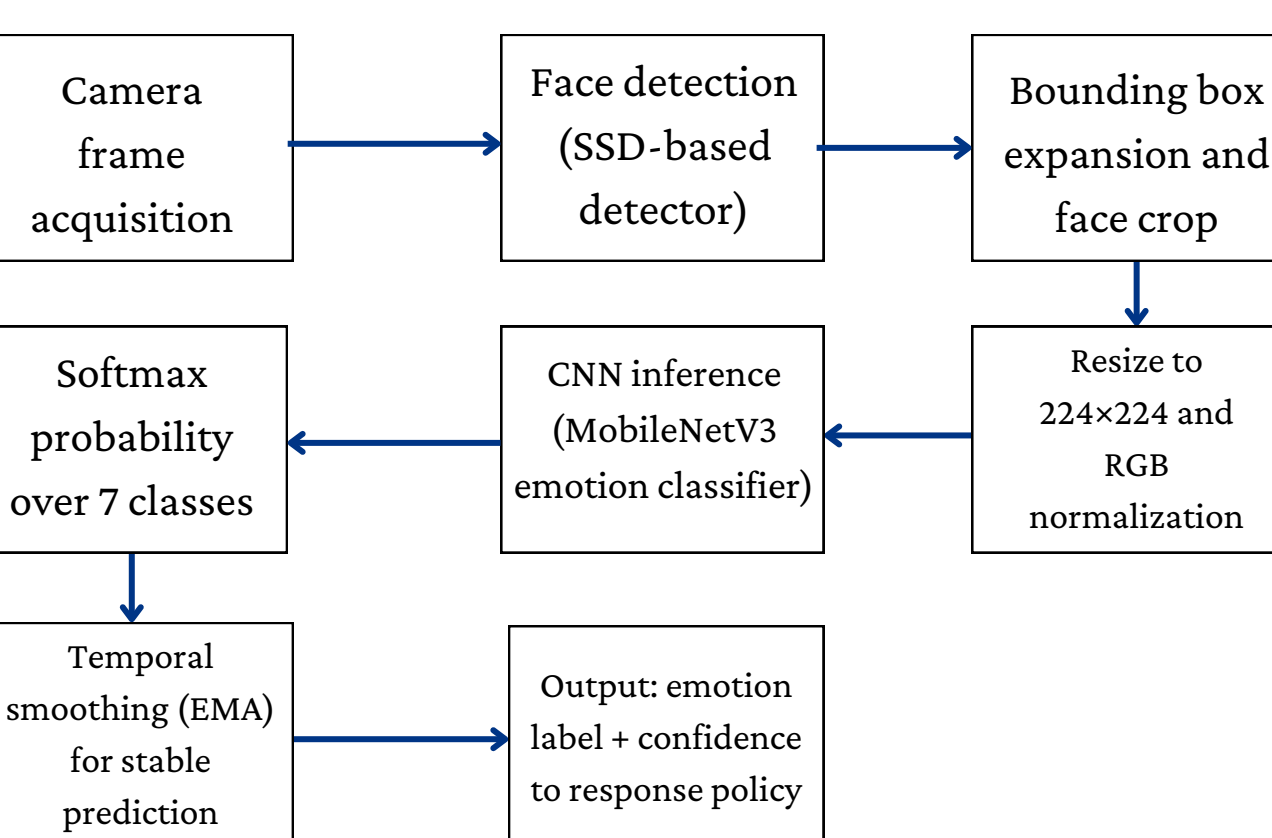


Figure 1 : Arousal-valence image for face detection

Mathematically, the point is computed as a weighted average: Valence Estimate and Arousal Estimate, where p_k is the predicted probability of class k . In real-time use, smoothing is applied so the point moves stably on the map instead of jumping abruptly between emotions.

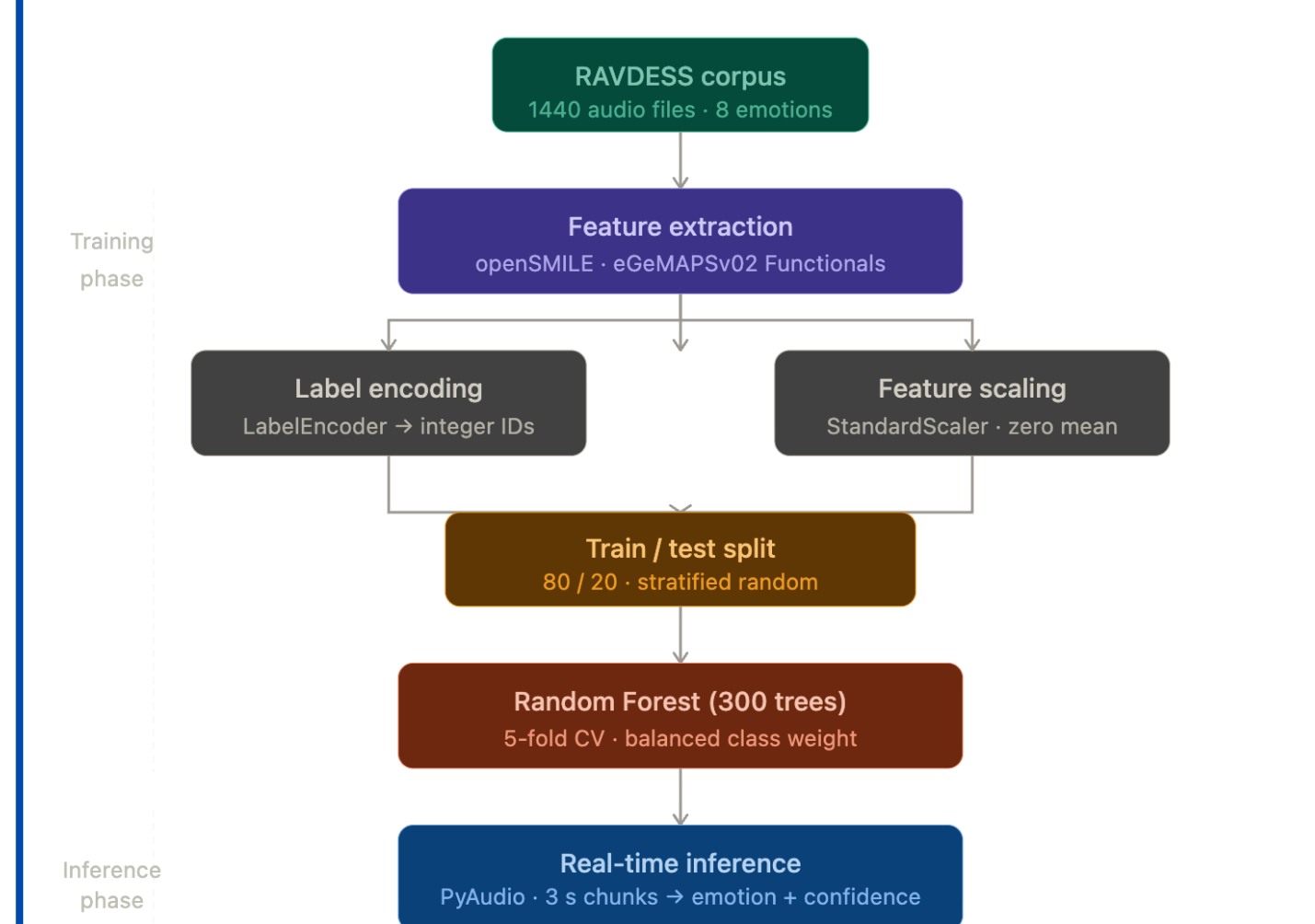
- 1) Class Probabilities
$$p_k = P(x_k | x), \quad \sum_{k=1}^K p_k = 1$$
- 2) Valence Estimate
$$v = \sum_{k=1}^K p_k v_k$$
- 3) Arousal Estimate
$$a = \sum_{k=1}^K p_k a_k$$
- 4) Confidence
$$c = \max_k p_k$$
- 5) Final Emotion Class
$$\hat{y} = \arg \max_k p_k$$
- 6) Temporal Smoothing (EMA)
$$\hat{a}_t = \alpha \hat{a}_{t-1} + (1 - \alpha) \hat{a}_t$$

$$\hat{v}_t = \left[\hat{v}_t \right], \quad 0 < \alpha < 1$$

Figure 2: Equations for estimates and confidence

Speech Emotion Recognition

Speech emotion recognition uses vocal acoustics to classify a speaker's emotional state in real time. The system was trained on the RAVDESS corpus — 1,440 labeled audio files spanning 8 emotions — and deployed on-device to feed emotion predictions into the LLM dialogue policy.



- Dataset: RAVDESS - 1,440 files - 8 emotions (neutral, calm, happy, sad, angry, fearful, disgust, surprised)
- Features: openSMILE eGeMAPSv02 Functionals — captures prosody, spectral, and voice quality cues
- Model: Random Forest - 300 estimators - balanced class weights - 5-fold cross-validation
- Real-time: PyAudio stream → 3-second chunks → eGeMAPSv02 → emotion label + confidence score

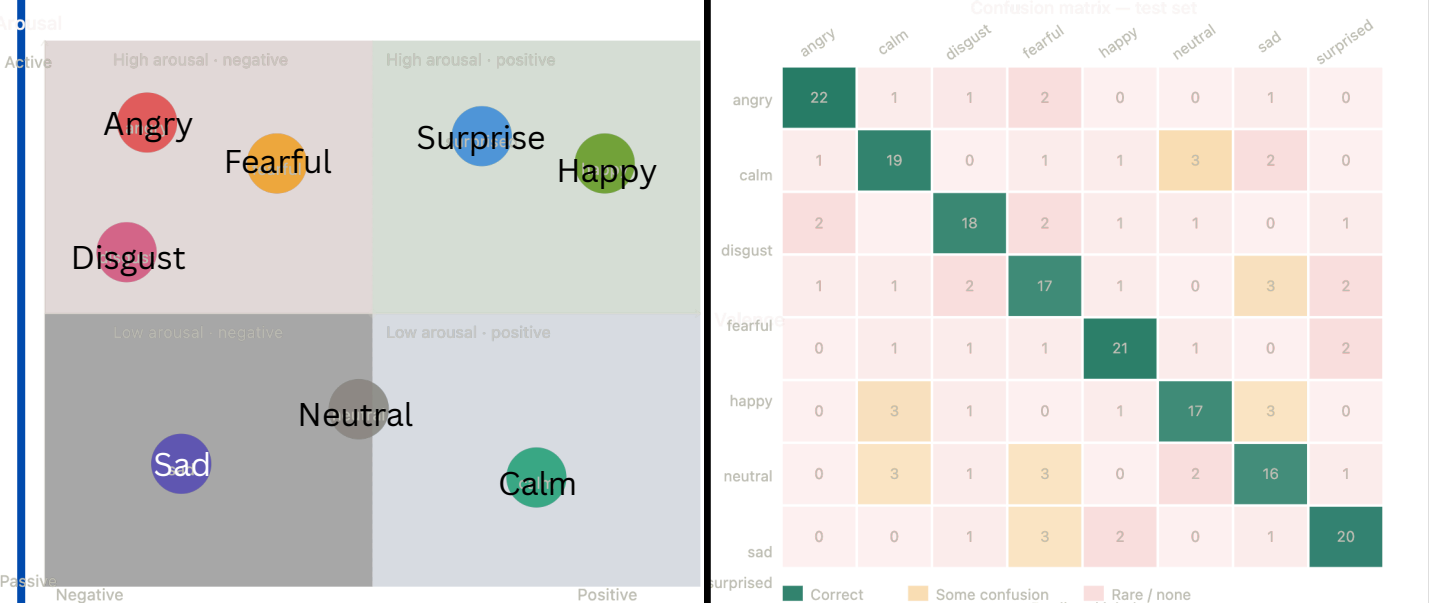


Figure 3 : Emotion compass for speech detection

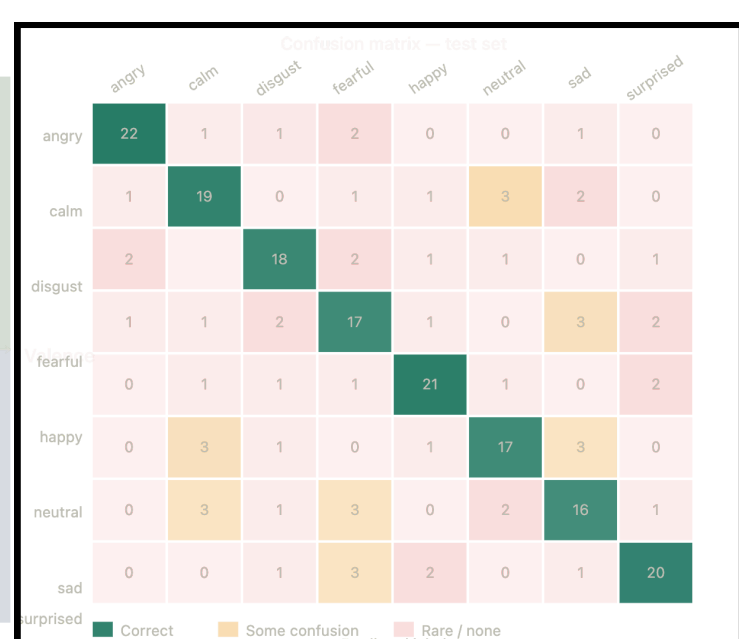
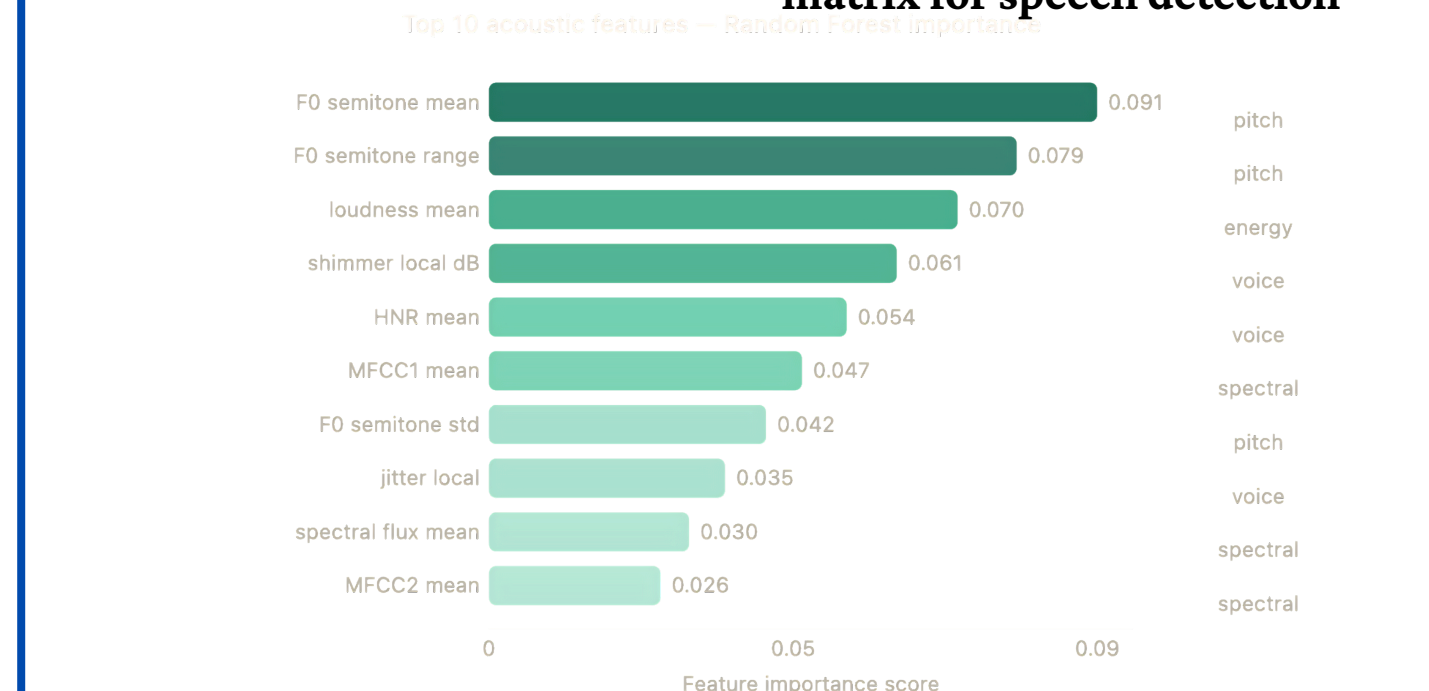


Figure 4 : Example of confusion matrix for speech detection



LLM model/Vibe Voice

This project utilized an open source version of Qwen3 as the base LLM for this project. The LLM was paired with the TTS software VibeVoice, which allows the LLM to take in both text and voice recorded responses from the user and respond in a text to speech voice.

The LLM takes in predictions and data sent from speech and facial recognition software to determine the user's current emotion. The LLM uses that prediction to formulate its response accordingly.

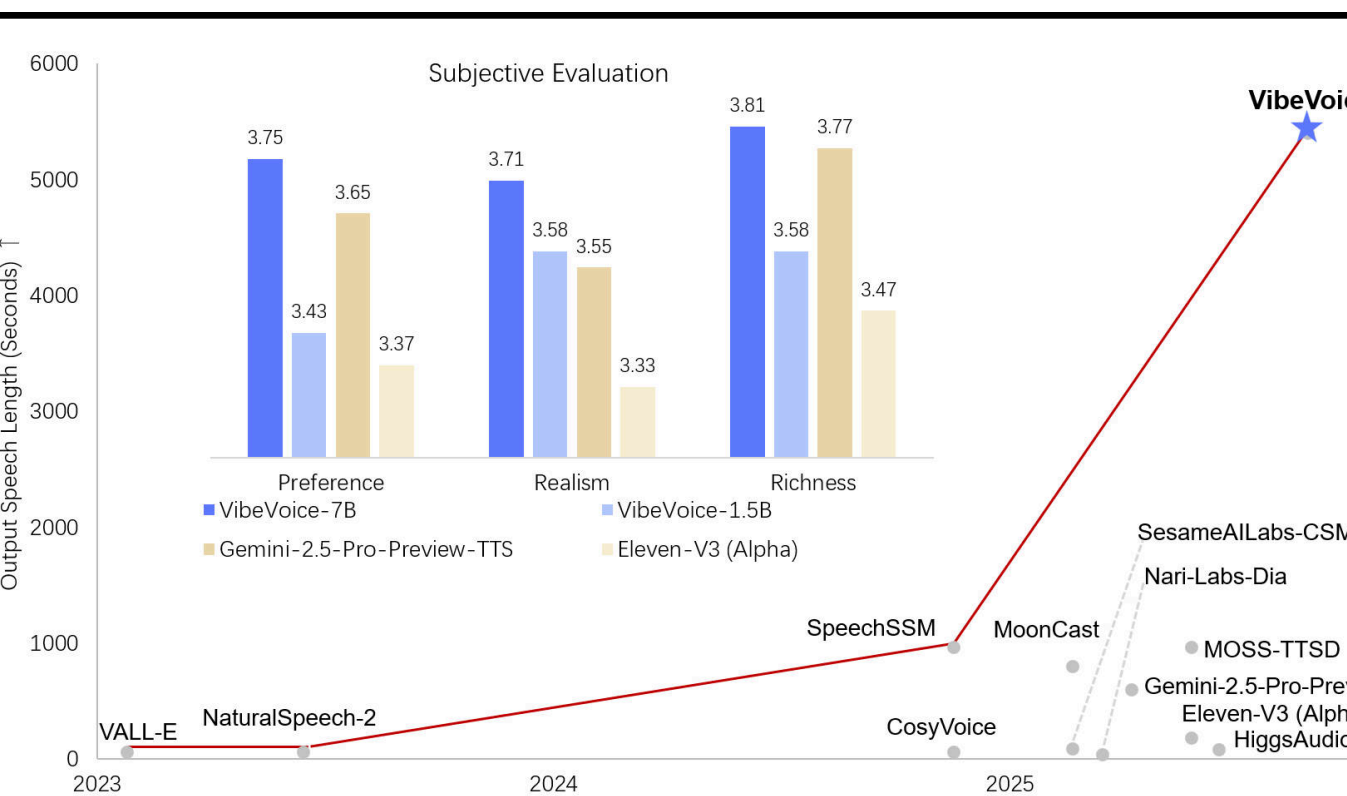


Figure 5: Subjective evaluation graphical comparison between different TTS models

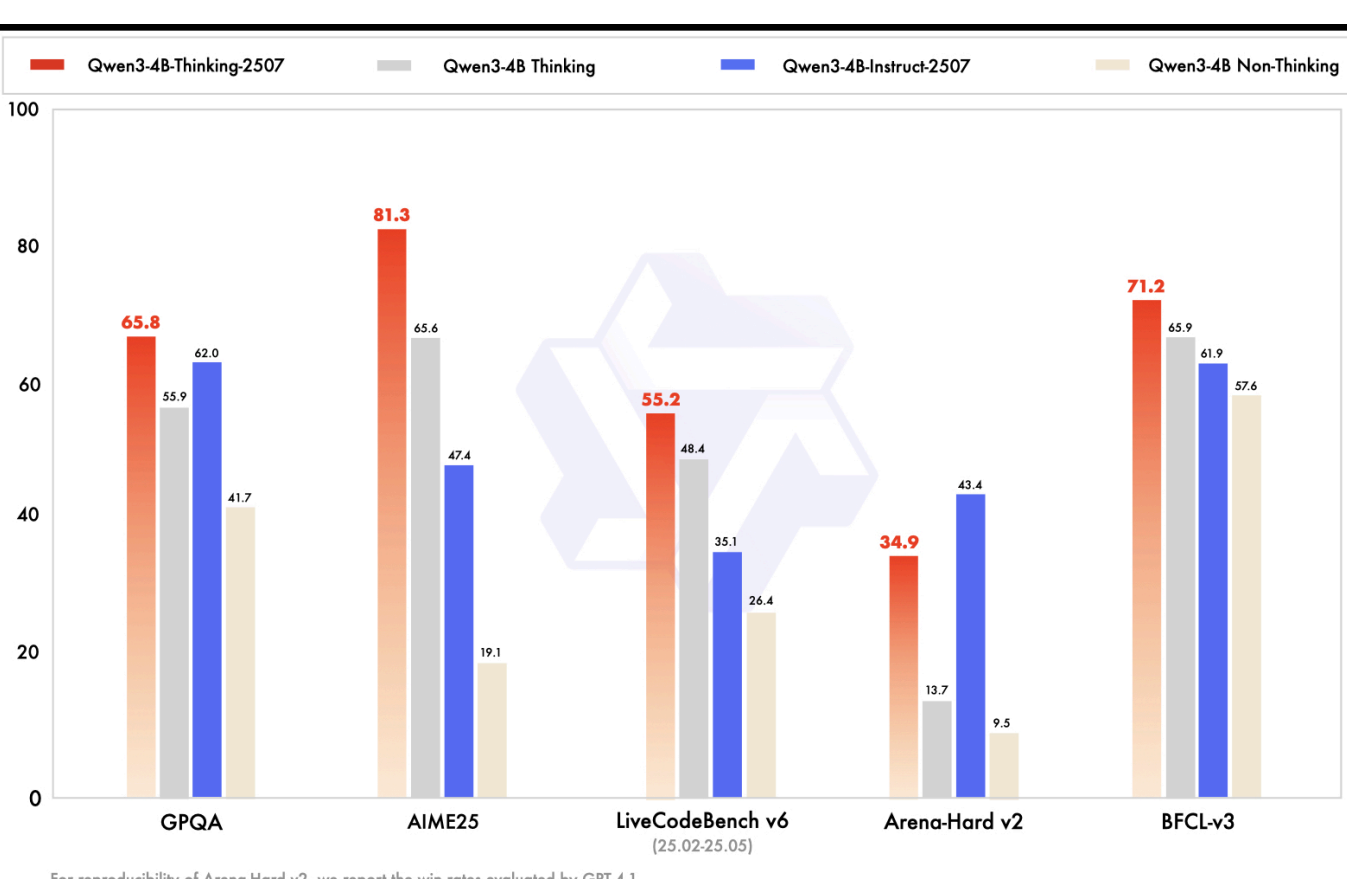
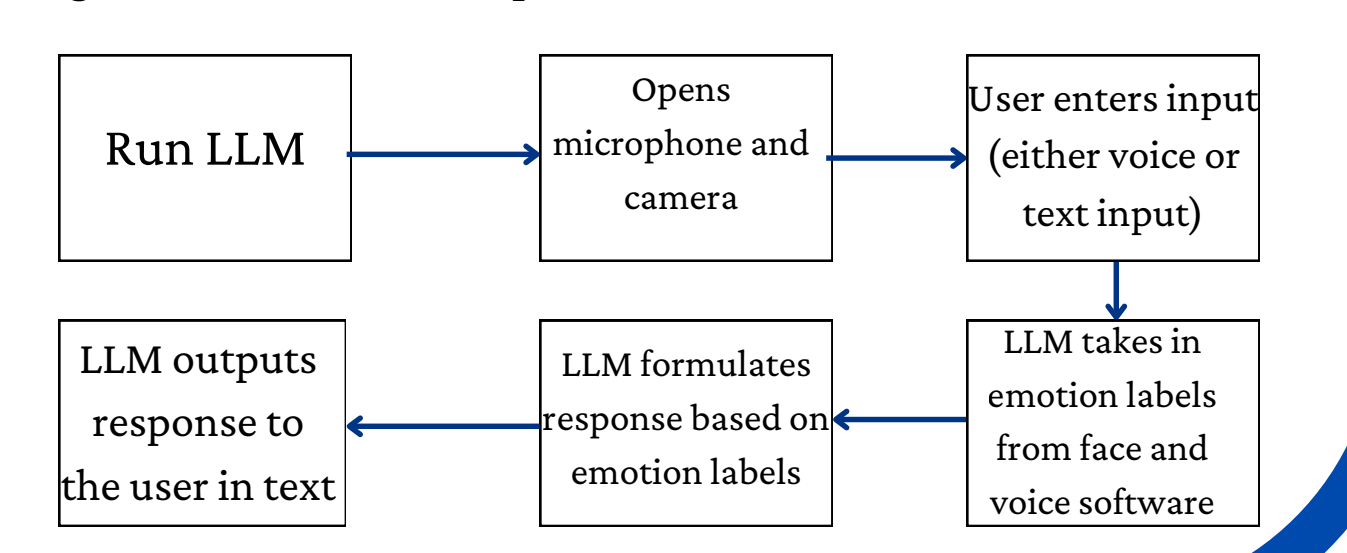


Figure 6: Performance comparisons between different Qwen3-4B models



Results

Facial Affect Sensing

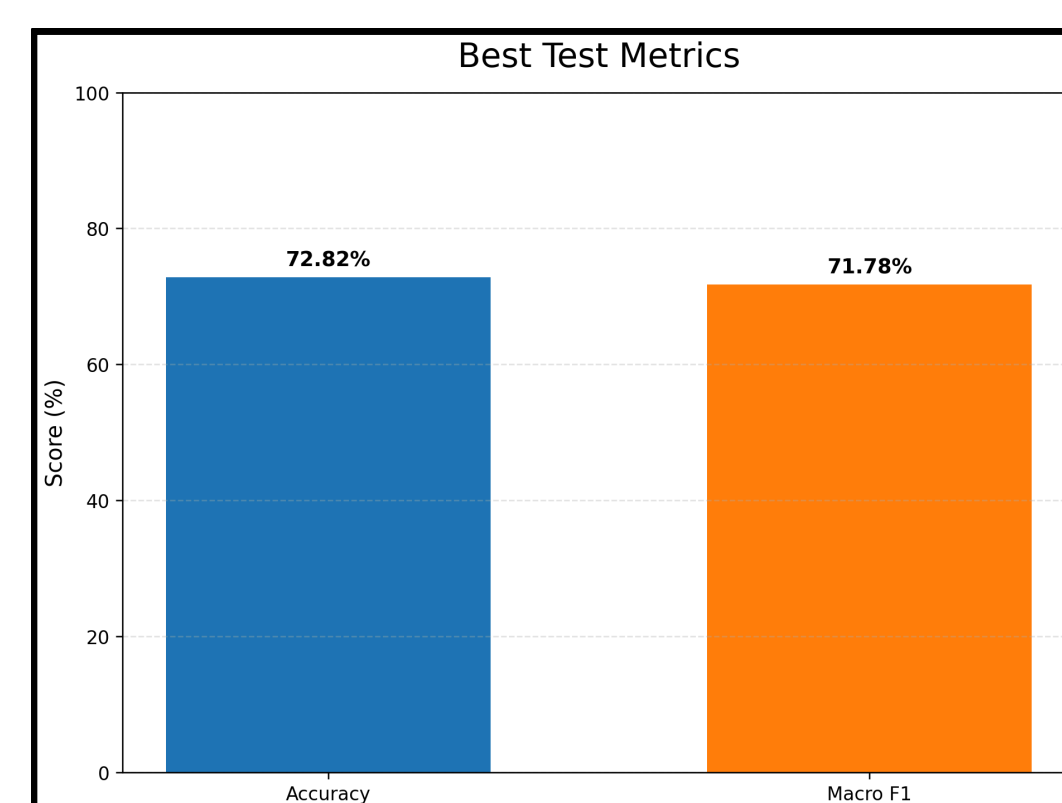


Figure 7. Overall test performance of the emotion recognition model (Face)

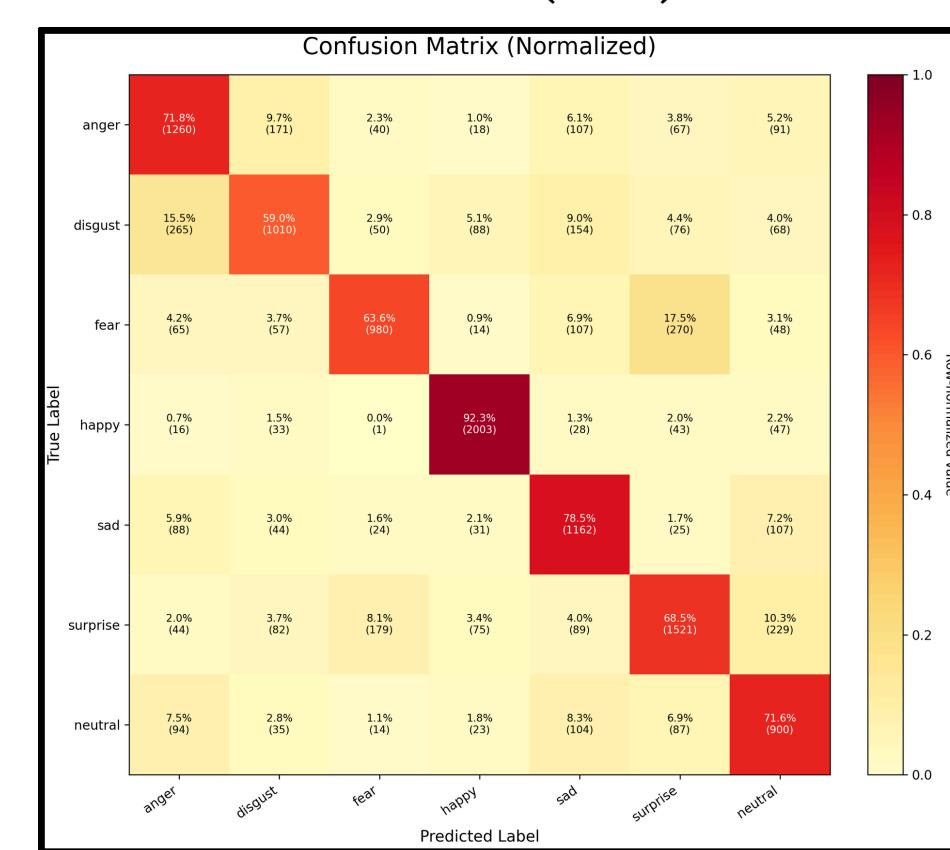


Figure 8. Normalized confusion matrix for test predictions (Face)

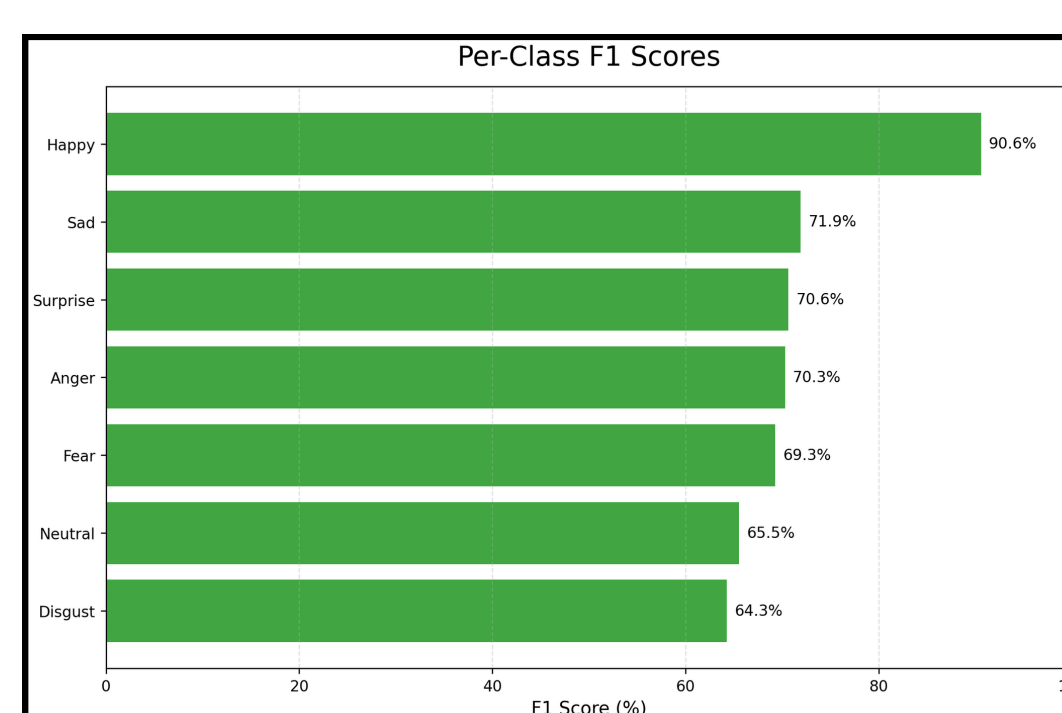


Figure 9. Per-class F1-scores on the test set (Face)

Speech Emotion Recognition

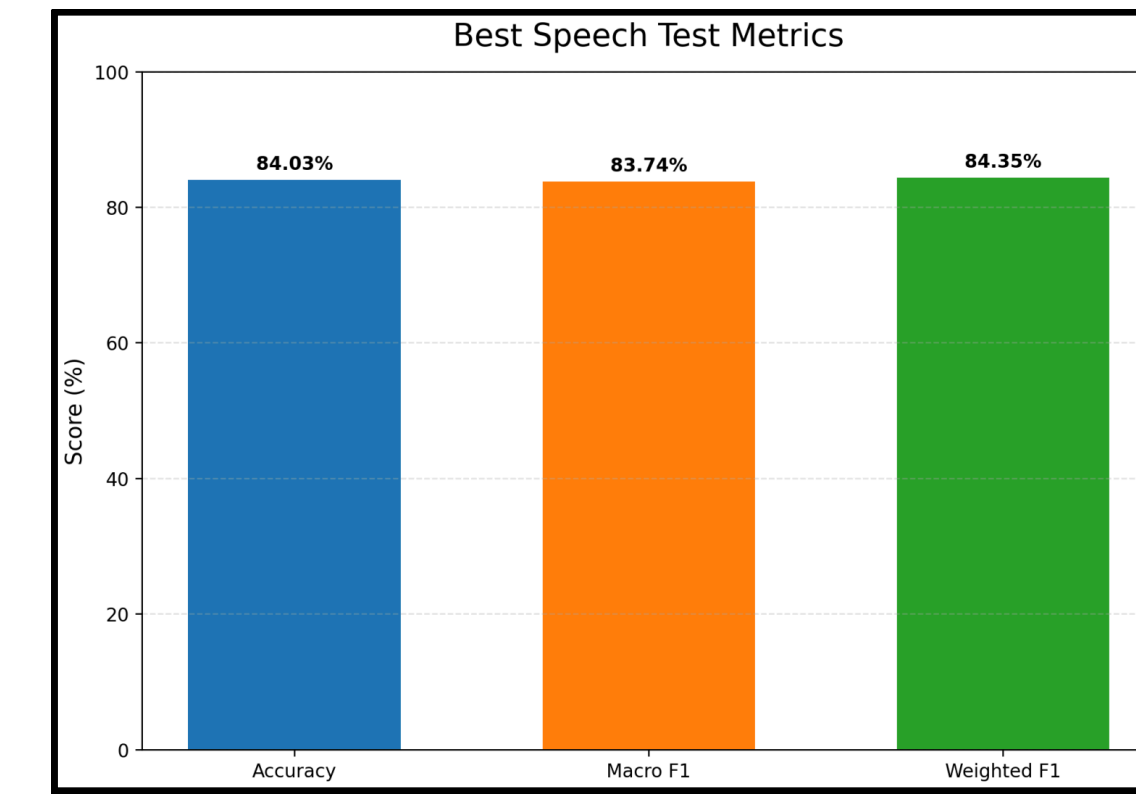


Figure 10. Overall test performance of the emotion recognition model (Speech)

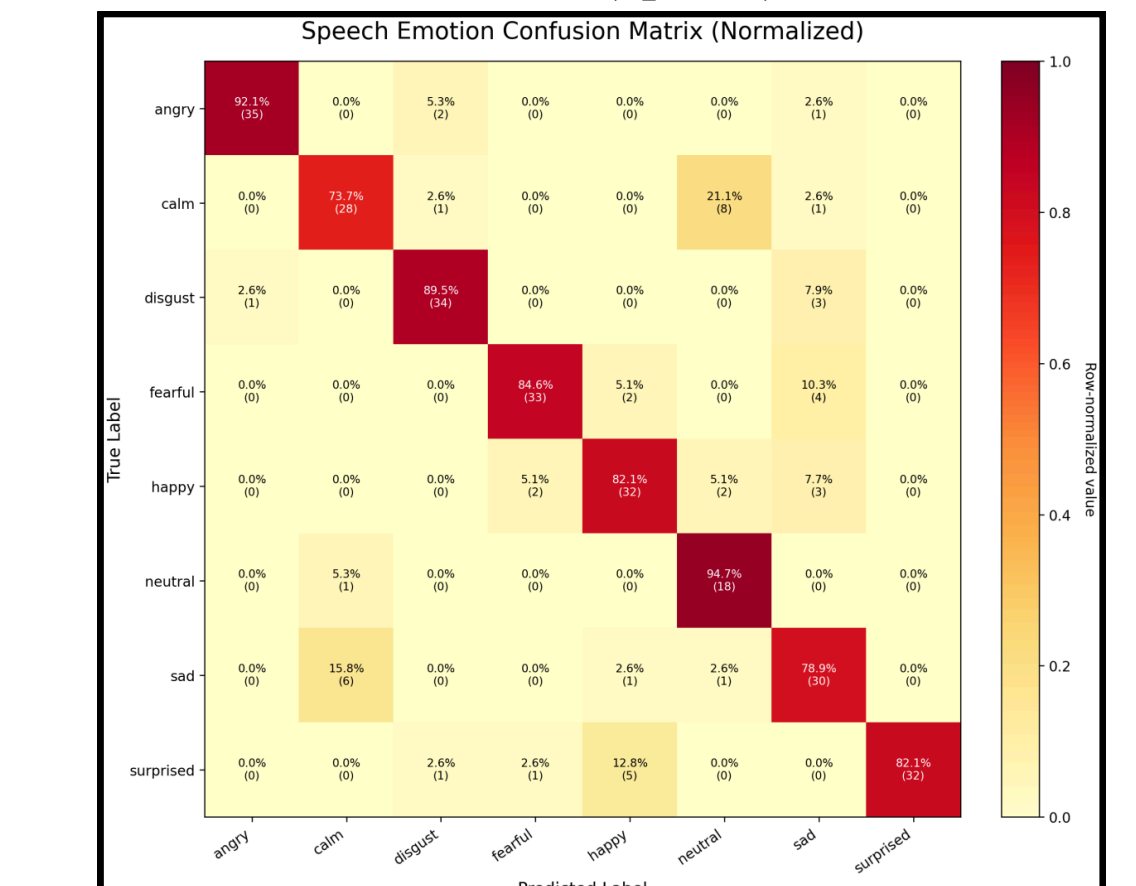


Figure 11. Normalized confusion matrix for test predictions (Speech)

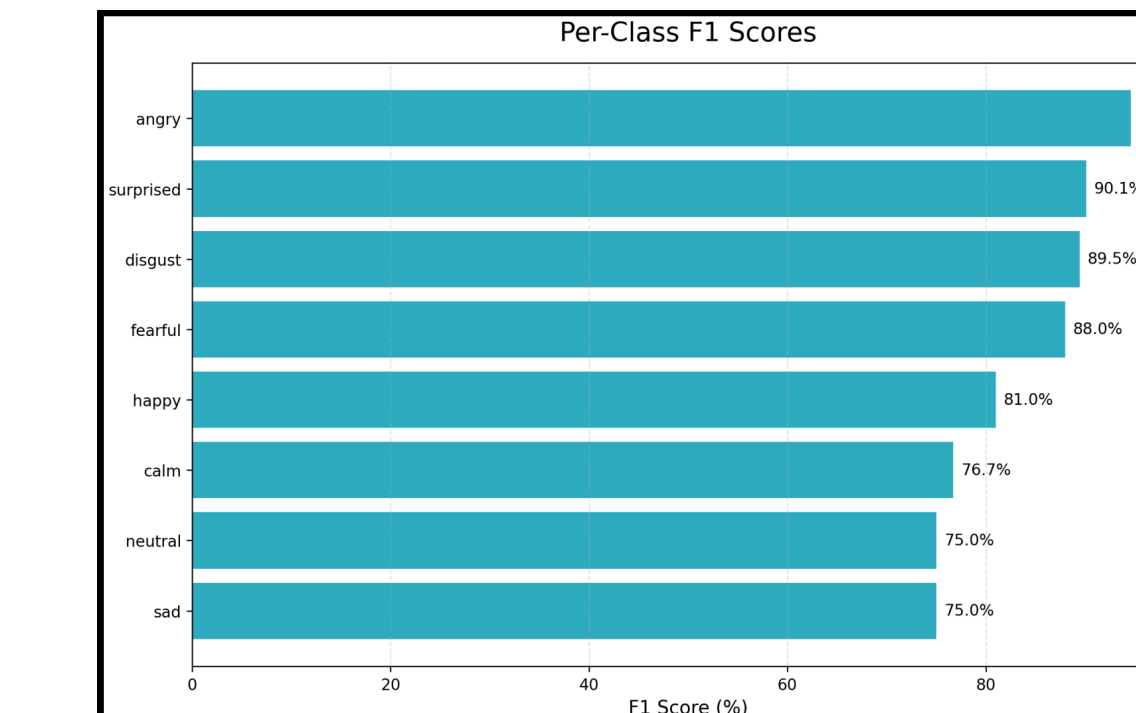


Figure 12. Per-class F1-scores on the test set (Speech)

Website for LLM models

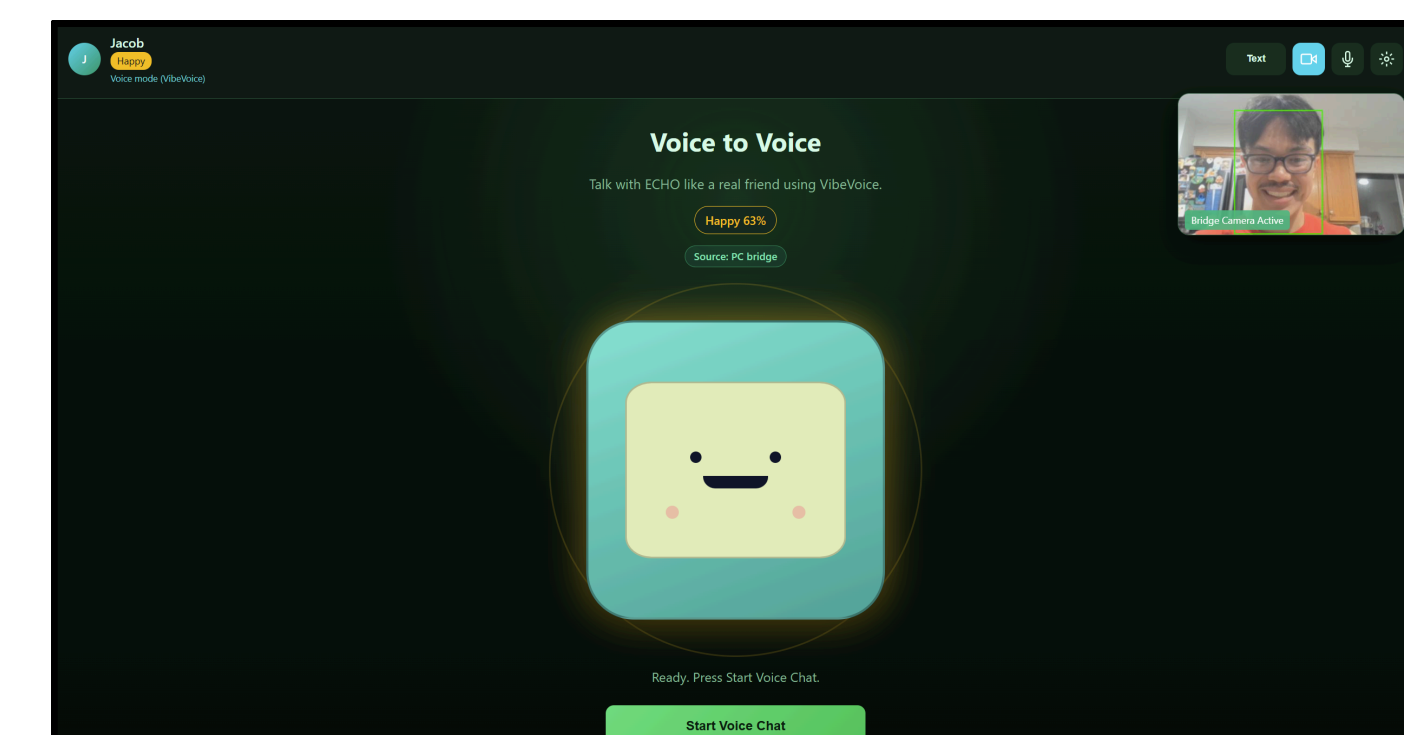


Figure 13: LLM Webpage for Voice to Voice Conversation

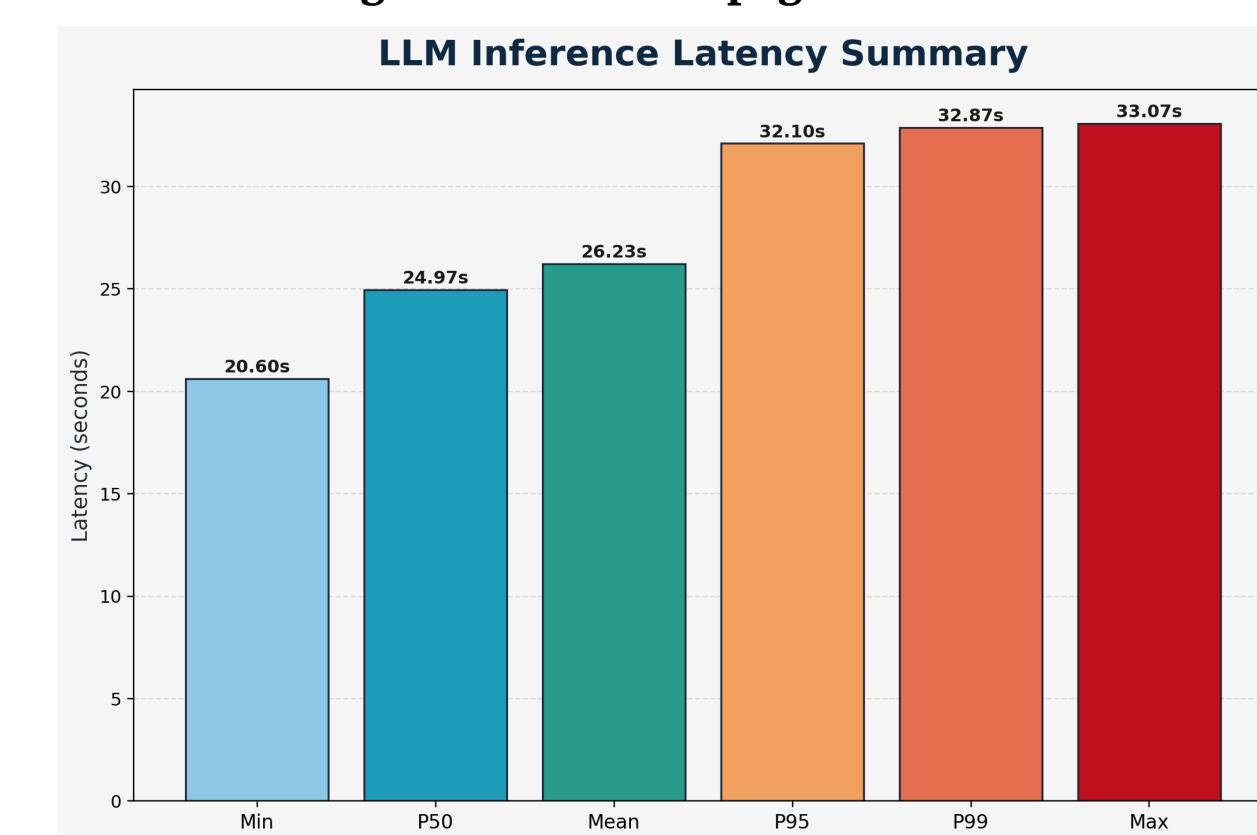


Figure 14: LLM Inference Latency Summary + Benchmark

Benchmark Details:
Device: CUDA
Runs: 10
Avg output tokens/run: 73.1
Throughput: 2.79
Tokens/sec
Model: Qwen/Qwen3-4B-Instruct-2507

Ultimately, the LLM was able to detect and make a prediction for all emotions. The LLM has its own webpage that can be used to access the LLM and its capabilities. It displays the current emotion that it thinks the user is experiencing and formulates all of its responses around that prediction. When using the text option for conversation, the LLM will leave a note along with the response telling the user the intended tone of voice it uses. This tone of voice changes as the LLM predicts what your emotion will be. The model will also display how confident it is in its prediction.

Overall, the results were a success and the LLM was able to make predictions on the user's emotions based on the inputs given to the model. This model can be improved upon and can be used in many different applications.

Conclusions

This project shows that combining real-time emotion sensing with an LLM can make human-AI interaction more context-aware and natural. By mapping facial affect signals into a continuous emotional state and conditioning response style on that state, the assistant can move beyond text-only replies toward communication that better matches user context. While the system does not “feel” emotion, it can learn emotion-related patterns and use them to improve perceived empathy, responsiveness, and conversational flow.

Possible Use Case:

For therapy-adjacent support, this framework could be used as a low-risk emotional check-in companion, stress-monitoring coach, or guided reflection tool between sessions, with clear boundaries that it is not a licensed therapist. For general LLM applications, the same approach can improve tutoring, elder care support, customer service, and accessibility interfaces by producing calmer, more supportive, and more adaptive responses in real time. Future work should include clinical collaboration, stronger safety guardrails, bias/fairness evaluation across demographics, and controlled user studies to validate impact.

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